A Practical Activity Report

for

MICROPROCESSOR BASED SYSTEM DESIGN

LAB ASSIGNMENT - II

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EXPERIMENT 1

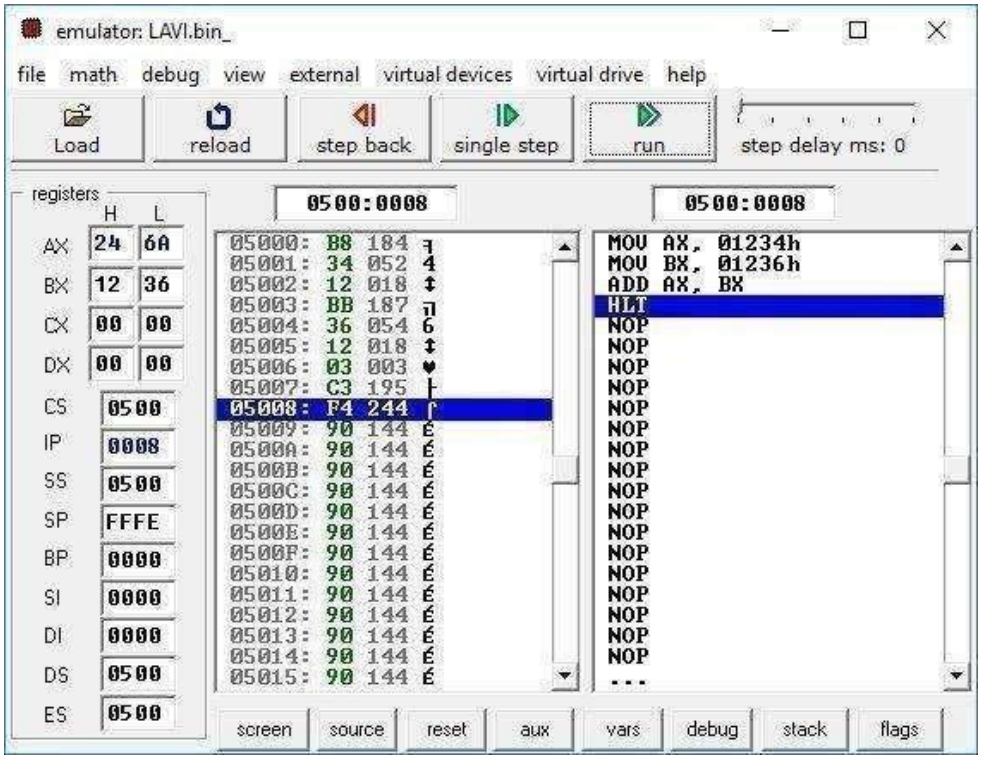
AIM:

Write an assembly language program to add two 16-bit numbers in 8086.

CODE:

```
MOV AX,1234H
MOV BX,1236H
ADD AX,BX
HLT
```

OUTPUT:



EXPERIMENT 2

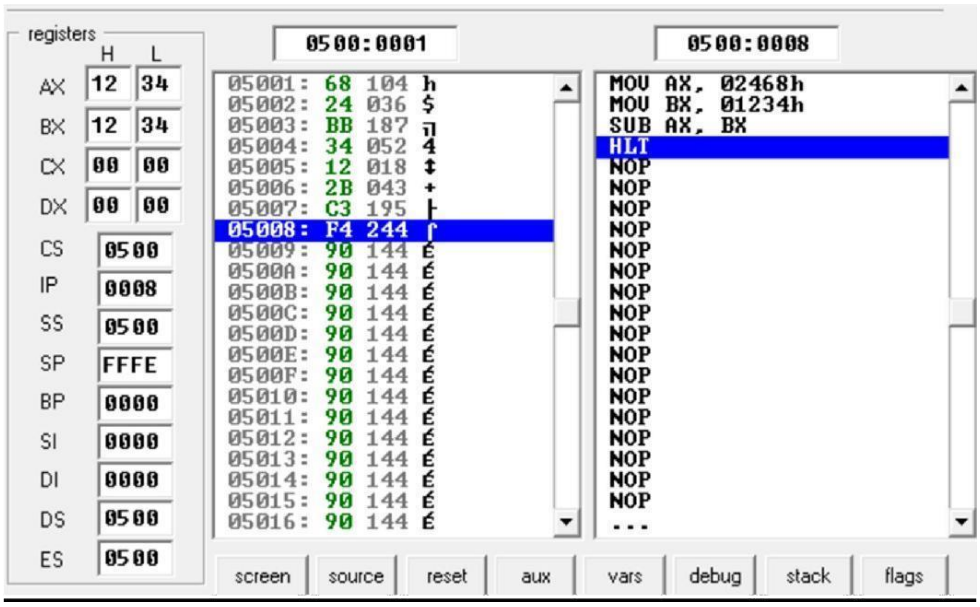
AIM:

Write an assembly language program to subtract two 16-bit numbers in 8086.

CODE:

```
MOV AX,2468H
MOV BX,1234H
SUB AX,BX
HLT
```

OUTPUT:



EXPERIMENT 3

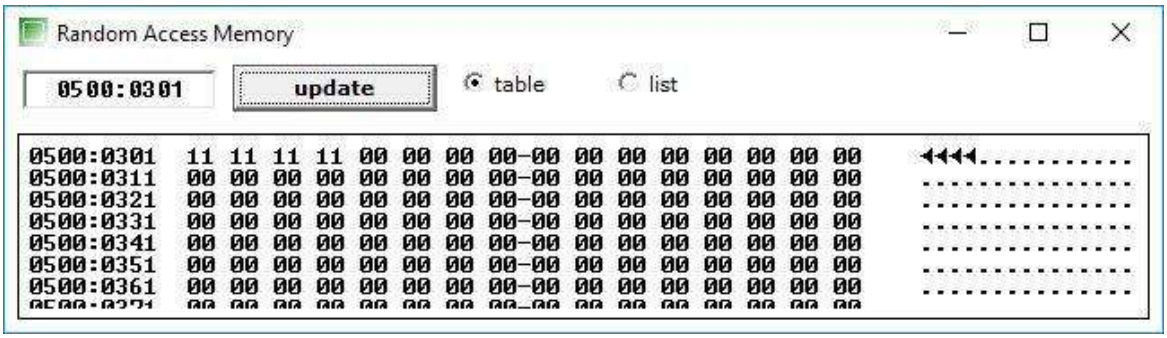
AIM:

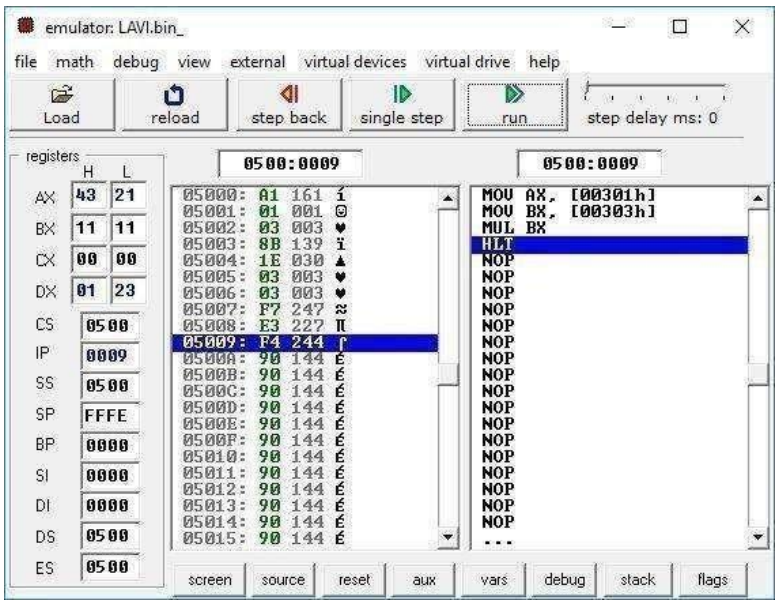
Write an assembly language program to multiply two 16-bit numbers in 8086.

CODE:

```
MOV  AX,[0301H]
MOV  BX,[0303H]
MUL BX
HLT
```

OUTPUT:





EXPERIMENT 4

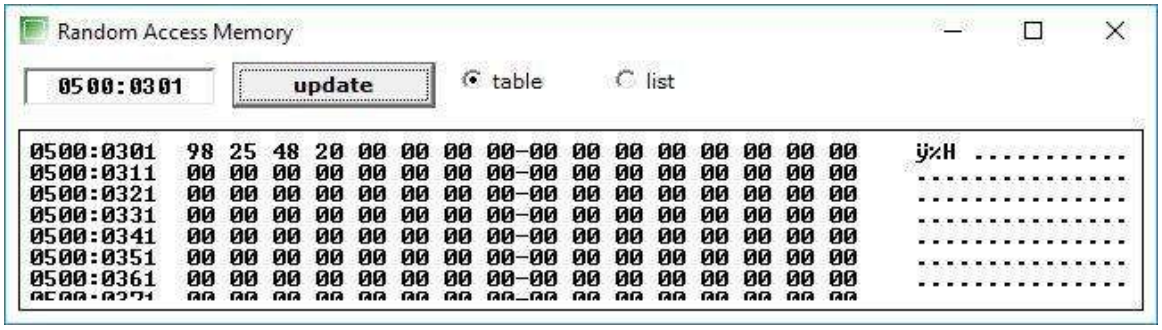
AIM:

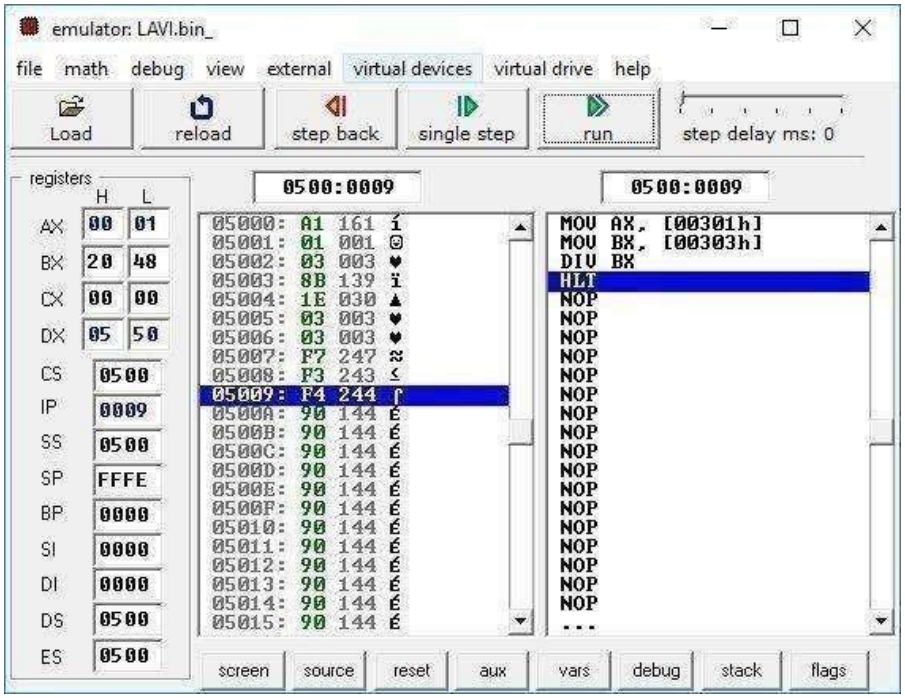
Write an assembly language program to divide two 16-bit numbers in 8086.

CODE:

```
MOV  AX,[0301H]
MOV  BX,[0303H]
DIV BX
HLT
```

OUTPUT:





EXPERIMENT 5

AIM: Write an assembly language program to demonstrate AAA, AAS, AAM, AAD, DAA and DAS in 8086.

CODE:

AAA	AAS

SUB AH,AH
MOV AL, '6'
ADD AL, '7'
AAA
OR AL, 30H
HLT
	SUB AH,AH
	MOV AL, '3'
	SUB AL, '9'
	AAS
	OR AL, 30H
	HLT

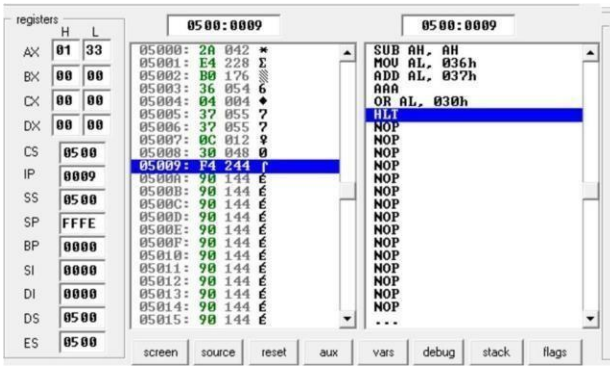
AAM	**AAD**

MOV AL, '3'
MOV BL, '9'
AND AL, 0FH
AND BL, 0FH
MUL BL
AAM
OR AL, 30H
HLT
	MOV AX, 0207H
	MOV BL, 05H
	AAD
	DIV BL
	HLT

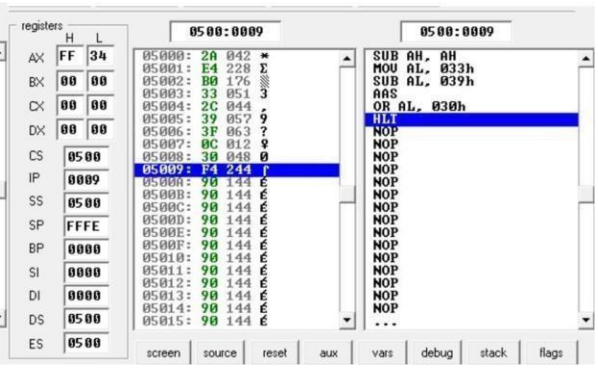
DAA	**DAS**

MOV AL, 71H
ADD AL, 43H
DAA
HLT
	MOV AL, 71H
	SUB AL, 43H
	DAS
	HLT

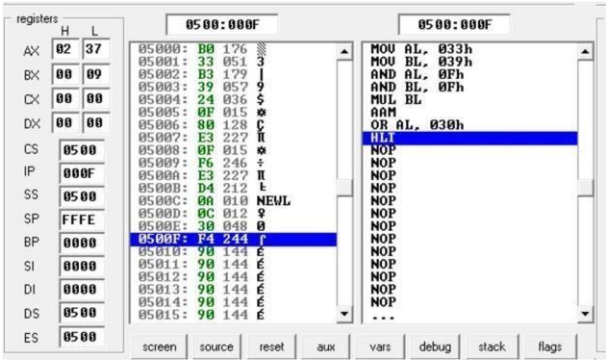
OUTPUT:



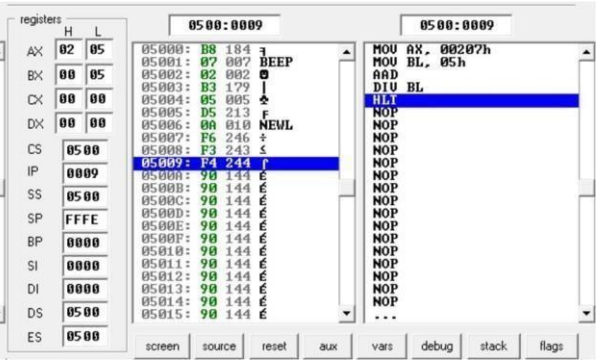
(AAA)



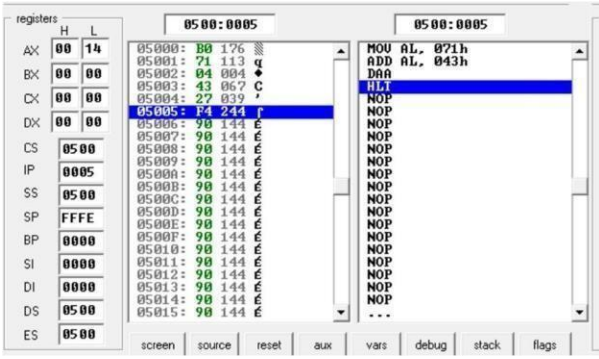
(AAS)



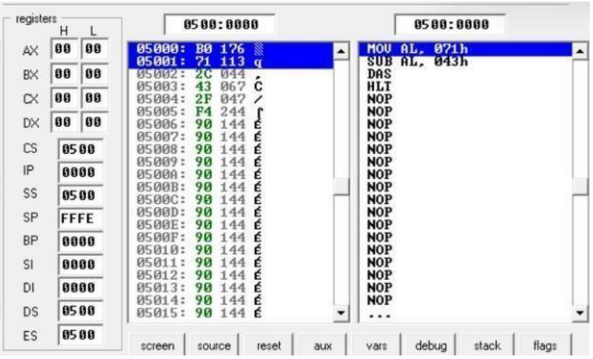
(AAM)



(AAD)



(DAA)



(DAS)

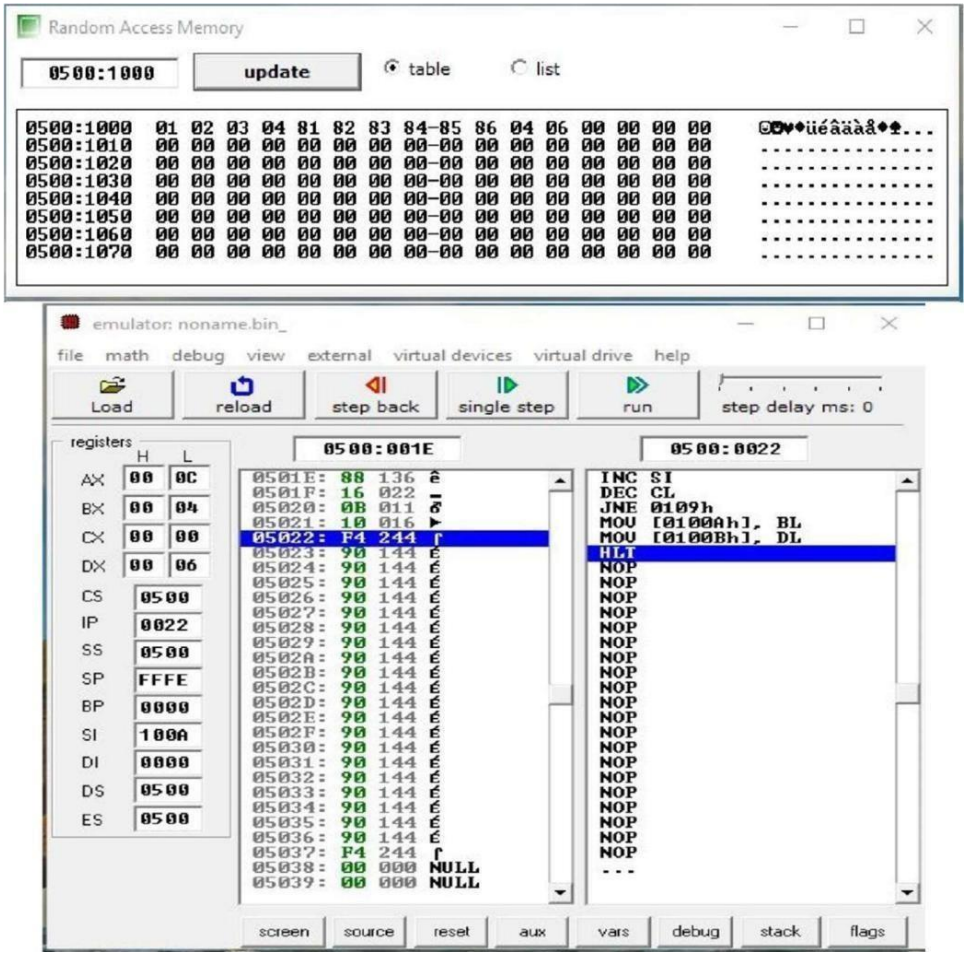
EXPERIMENT 6

Aim: Write an assembly language program to find out the count of positive numbers and negative numbers from a series of signed numbers in 8086.

CODE:

```
MOV CL,0AH
MOV BL,00H
MOV DL,00H
LEA SI, [1000H]
L1: MOV AL, [SI]
SHL AL, 01
JNC L2
INC DL
JMP L3
L2: INC BL
L3: INC SI
DEC CL
JNZ L1
MOV [100AH], BL
MOV [100BH], DL
HLT
```

OUTPUT:


Random Access Memory
0500:1000
update
table
list
0500:1000 01 02 03 04 81 82 83 84-85 86 04 06 00 00 00 00
0500:1010 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0500:1020 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0500:1030 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0500:1040 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0500:1050 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0500:1060 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0500:1070 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
emulator: noname.bin_
file math debug view external virtual devices virtual drive help
Load
reload
step back
single step
run
step delay ms: 0
registers
H L
AX 00 0C
BX 00 04
CX 00 00
DX 00 06
CS 0500
IP 0022
SS 0500
SP FFFE
BP 0000
SI 100A
DI 0000
DS 0500
ES 0500
0500:001E
0501E: 88 136 ê
0501F: 16 022 ▬
05020: 0B 011 ♂
05021: 10 016 ►
05022: F4 244 ⌠
05023: 90 144 É
05024: 90 144 É
05025: 90 144 É
05026: 90 144 É
05027: 90 144 É
05028: 90 144 É
05029: 90 144 É
0502A: 90 144 É
0502B: 90 144 É
0502C: 90 144 É
0502D: 90 144 É
0502E: 90 144 É
0502F: 90 144 É
05030: 90 144 É
05031: 90 144 É
05032: 90 144 É
05033: 90 144 É
05034: 90 144 É
05035: 90 144 É
05036: 90 144 É
05037: F4 244 ⌠
05038: 00 000 NULL
05039: 00 000 NULL
0500:0022
INC SI
DEC CL
JNE 0109h
MOV [0100Ah], BL
MOV [0100Bh], DL
HLT
NOP
NOP
NOP

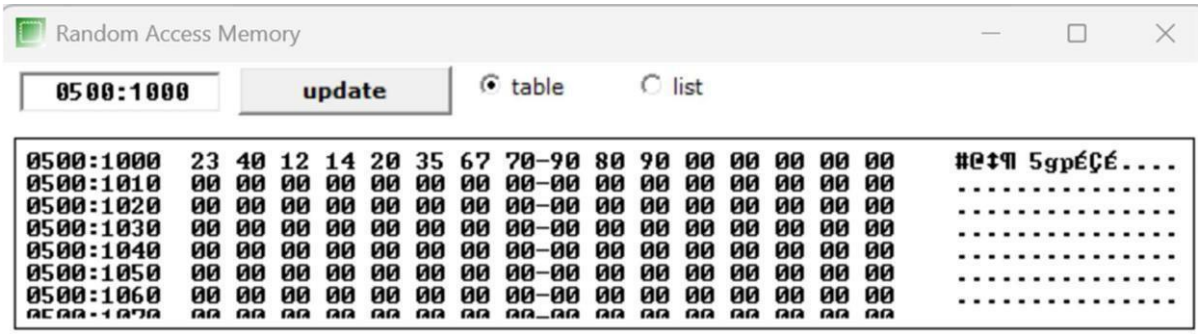
EXPERIMENT 7

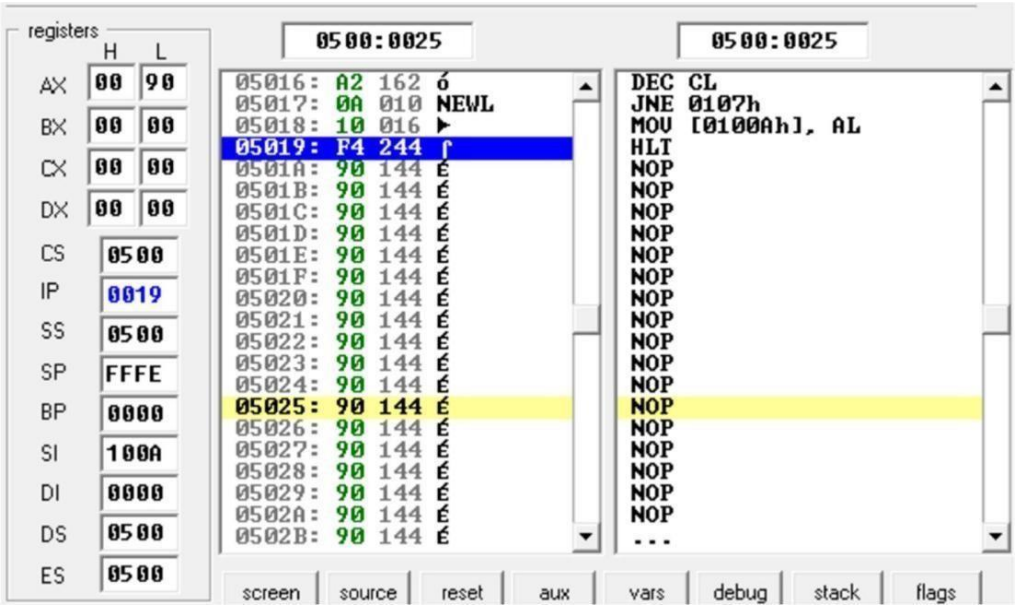
Aim: Write an assembly language program to convert to find out the largest number from a given unordered array of 8-bit numbers, stored in the locations starting from a known address in 8086.

CODE:

```
MOV BX, 1000H
MOV CL, [BX]
INC BX
MOV AX, [BX]
DEC CL
Back: INC BX
INC BX
CMP AX, [BX]13
JNC Next
MOV AX, [BX]
Next: DEC CL
JNZ Back
MOV [1020H], AX
HLT
```

OUTPUT:





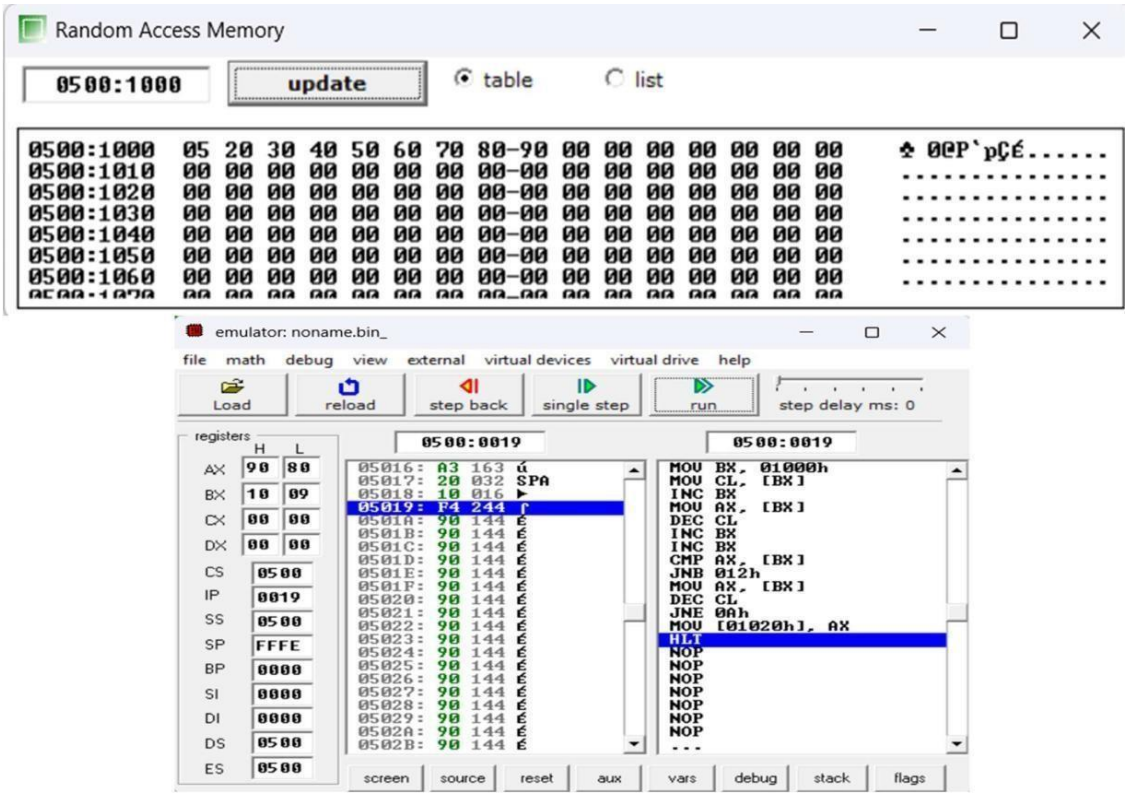
EXPERIMENT 8

Aim: Write an assembly language program to find out the largest number from a given unordered array of 16-bit numbers, stored in the locations starting from a known address in 8086.

CODE:

```
MOV BX, 1000H
MOV CL, [BX]
INC BX
MOV AX, [BX]
DEC CL
Back: INC BX
INC BX
CMP AX, [BX]
JNC Next
MOV AX, [BX]
Next: DEC CL
JNZ Back
MOV [1020H], AX
HLT
```

OUTPUT:



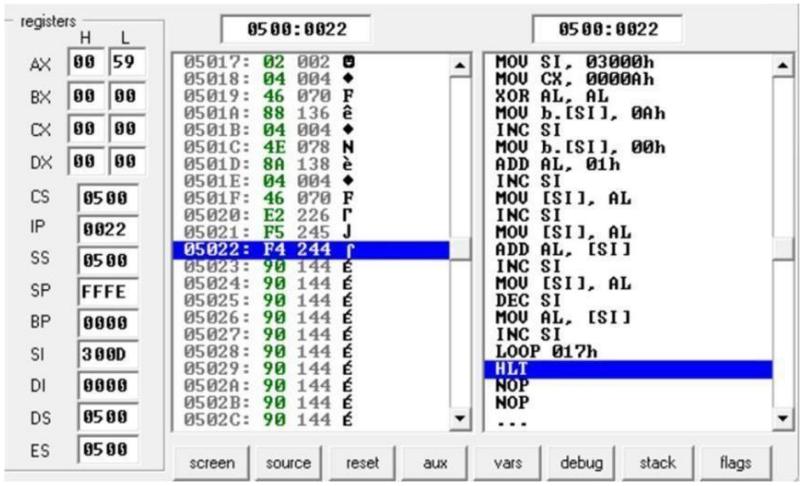
EXPERIMENT 9

Aim: Write an assembly language program to print Fibonacci series in 8086.

CODE:

```
MOV SI,3000H
MOV CX,0AH
XOR AL, AL
MOV [SI],0AH
INC SI
MOV  [SI],00H
ADD  AL,  01H
INC SI
MOV  [SI],  AL
INC SI
MOV [SI],AL
Back: ADD AL,[SI]
INC SI
MOV [SI], AL
DEC SI
MOV AL, [SI]
INC SI
LOOP Back
HLT
```

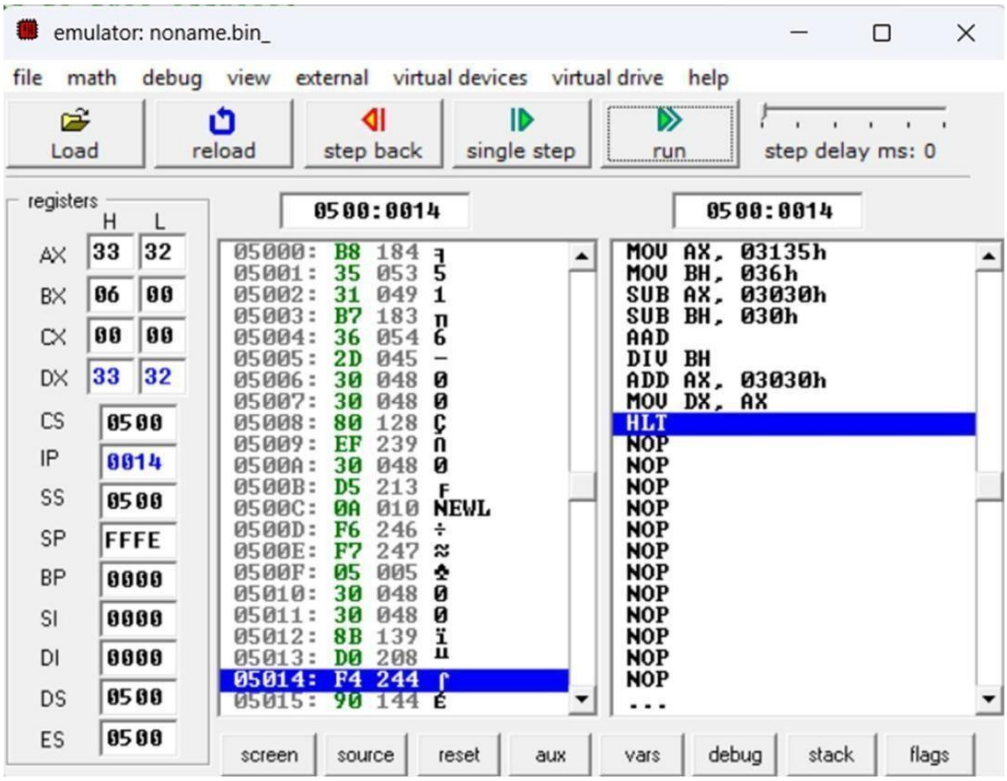
OUTPUT:



EXPERIMENT 10

Aim: Write an assembly language program to perform the division 15/6 using the ASCII codes. Store the ASCII codes of the result in register DX.

CODE:

```
MOV AX,'15'
MOV BH,'6
SUB AX,3030H
SUB BH,30H
AAD
DIV BH
ADD AX,3030H
MOV DX, AX
HLT
```

OUTPUT: